

RESEARCH ARTICLE

Advancing Aquatic eDNA Applications for Rare Species

Maximizing eDNA detections of rare marine organisms: A case study of a delphinid species

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Abstract

1. Environmental DNA (eDNA) is an emerging tool for surveying marine organisms. Advances in molecular assay designs and bioinformatic methods provide new opportunities for detecting rare species; however, overcoming methodological challenges of rare species detection remains an important obstacle for the widespread deployment of eDNA methods for biomonitoring.
2. Here, we provide a rigorous evaluation of eDNA sampling and processing methods to optimize eDNA capture and detection of a rare delphinid species. We applied a hierarchical Bayesian occupancy model to separate the effects of presence, capture and detection of target eDNA from *Delphinus delphis* and its two associated subspecies. We used a highly replicated sampling effort in the California Current Ecosystem to assess (1) the effect of water depth on the probability of *D. delphis* eDNA presence, (2) the effects of filtered water volume and filtration method on the likelihood of capturing *D. delphis* eDNA given its presence and (3) the effect of primer choice on the probability of *D. delphis* eDNA detection given its capture.
3. Our results show that *D. delphis* eDNA is most likely to occur in surface (0–10 m) waters, and that large volume (>4 L) sampling increased the probability of capturing *D. delphis* eDNA on a filter. Moreover, the use of primers designed for cetaceans increased detection probabilities relative to more general primers designed for vertebrates.

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4. Our intensive replication in field sampling and laboratory assays and subsequent hierarchical modelling indicates that *D. delphis* eDNA was present in 90% of surface samples. Yet, since it makes up a tiny fraction of the ambient DNA pool, reliable detection requires optimized experimental design at both the sampling and processing stages. The modelling approach applied here provides an experimental design framework for eDNA studies focused on capturing rare targets. We use the results of this application to provide specific guidance on optimal experimental design for detection of delphinid eDNA.

KEYWORDS

dolphins, metabarcoding, probabilistic model, rare targets

1 | INTRODUCTION

Environmental DNA (eDNA) from seawater samples offers a scalable, noninvasive and sensitive method for detecting marine organisms (Deiner et al., 2021; Pikitch, 2018; Rodríguez-Ezpeleta et al., 2021). Seawater can be collected and filtered using a variety of commercially available and bespoke methods (Patin & Goodwin, 2023), making it feasible to collect eDNA samples from a range of research platforms. The versatility of eDNA sampling, combined with its taxonomic specificity and accuracy, is promising as a stand-alone or supplementary method for surveying rare species (Jerde et al., 2011; Pikitch, 2018), such as marine mammals (Suarez-Bregua et al., 2022). While uncertainty remains around the rates of eDNA decay and transport, there is growing evidence that the half-life of eDNA in the ocean is on the order of 6–24 h (Collins et al., 2018; Jo et al., 2020; Sriver et al., 2023; Brandão-Dias et al., 2025) with factors like temperature, starting amount and target gene length affecting precise degradation rates. Transport is highly dependent on oceanography (Andruszkiewicz et al., 2019; Xiong et al., 2025) but most studies estimate that eDNA signal in the open ocean is lost within relatively small (1–10 km) horizontal distances (Allan et al., 2021; Ellis et al., 2022; Jo et al., 2025; Shea et al., 2022). Vertical transport also does not seem to be a major factor in eDNA transport, with models suggesting a minor role for physical transport processes like sinking (Allan et al., 2021) and detections typically consistent with the known depth distribution of the target organism (Minamoto et al., 2017; Murakami et al., 2019). Positive eDNA detection is therefore a reliable indicator of recent organismal presence at a given location with great potential to improve surveys of rare organisms over large spatial scales.

Expanding the use of eDNA-based surveys has numerous implications for marine mammals, including cetaceans, which face increasing pressure from fishing, pollution, ship strikes, excessive noise and changing oceanographic conditions (Hazen et al., 2013; De Vere et al., 2018; US Department of the Navy, 2025). Contemporary management of cetaceans, mandated in the United States by the Marine Mammal Protection Act and Endangered Species Act, relies heavily on visual or acoustic observations to inform accurate

estimates of abundance and distribution (Abrahms et al., 2019; Becker et al., 2017, 2022). However, detection data are difficult and expensive to obtain due to the relatively rare and sometimes elusive nature of the target taxa in combination with often inclement survey conditions, especially during non-summer months (Becker et al., 2014; Kanaji et al., 2018). Moreover, some cetacean species are difficult to visually identify at sea or classify in passive acoustic recordings, resulting in poorly resolved data that can prevent accurate assessments of biogeography (Forney & Barlow, 1998; Carretta et al., 2011; Palacios et al., 2023; Becker et al., 2014; Becker et al., 2017). Estimates of abundance and distribution are highly uncertain for many cetacean species, even in geographic regions with extensive survey effort, such as the California Current Ecosystem (CCE; e.g. Kanaji et al., 2018; Carretta et al., 2024). The development of more comprehensive and taxonomically precise survey methods, when applied in combination with existing approaches, could inform our understanding of cetacean spatial ecology and response to environmental variability.

In the ocean, it is difficult to consistently detect eDNA of rare target organisms, such as cetaceans. Here, we define 'rare' organisms as those that are orders of magnitude lower in number relative to ubiquitous marine organisms like microbial plankton. For instance, nano- and picoplankton are measured at cells per mL, micro- and zooplankton in cells per L or per m³, and cetaceans as the number of individuals per km². Thus, relative to other marine organisms, cetaceans are low in number, widely dispersed and swim in a 'microbial soup' of bacteria, protists and plankton that dominate marine eDNA pools (Hatton et al., 2021; Patin et al., 2025; Pinfield et al., 2019; Robinson et al., 2023; Tringe et al., 2005). One study estimated the average percentage of cetacean eDNA in a marine water column sample at less than 0.0003% (Patin et al., 2025). This disparity directly impacts the likelihood of observation in molecular surveys, which depend not only on the presence of target organism DNA but also on the abundance of all other eDNA sources. The challenges associated with rare target detection stem from the fact that not only are the target organisms unlikely to have recently been physically present at any given location, but the amount of biological material (including DNA) they shed is small relative to other sources.

The high fraction of off-target DNA molecules found in a typical ocean water sample is a particular impediment for marker gene sequencing, or metabarcoding (Collins et al., 2019; Székely et al., 2022). In this approach, taxonomically broad primers can capture eDNA from dozens to thousands of species of interest in a single assay, making it appealing for biomonitoring applications (Andruszkiewicz et al., 2017; Closek et al., 2019; Klymus et al., 2017; Miya et al., 2020; Port et al., 2016; Valsecchi et al., 2020). Previous studies comparing eDNA with visual and acoustic survey methods for cetaceans have reported high false negative rates of eDNA-based detection (Pinfield et al., 2019; Robinson et al., 2023), reflecting the challenges with capturing and amplifying these rare molecular targets. Species-specific assays like quantitative and digital PCR are highly sensitive and have been successful at consistently detecting cetacean eDNA (Baker et al., 2018; Boldrocchi et al., 2024; Bylemans et al., 2019; Foote et al., 2012; Harper et al., 2018; Robinson et al., 2024), although this may reflect a publication bias for positive results as examples of qPCR failures also exist (e.g. Pinfield et al., 2019). However, the need for a unique assay design for each target species limits the scope of biodiversity data that can be generated (McCarthy et al., 2023). For more comprehensive biomonitoring, there is a pressing need to optimize and improve metabarcoding methods for reliable, accurate detection of multiple rare species.

Fortunately, multiple studies show the challenges associated with rare targets can be overcome with appropriate sampling and laboratory processing approaches (Duarte et al., 2023; Suarez-Bregua et al., 2022). For instance, Boyse et al. (2024), Deng et al. (2025), Juhel et al. (2021) and Nikolic et al. (2025) successfully surveyed cetaceans with metabarcoding by filtering large (10–30 L) water volumes. Nikolic et al. (2025) and Juhel et al. (2021) further applied 12 technical PCR replicates to every sample, while Alter et al. (2022) collected multiple 1-L biological replicates in direct proximity to cetaceans and applied ultra-deep sequencing to generate over 2 million reads per sample on average. These examples demonstrate the potential for efficient cetacean surveys using fit-for-purpose methods. Moreover, metabarcoding primers targeting taxon-specific genes are sensitive relative to more general primers (Deng et al., 2025; McCarthy et al., 2023), can differentiate among marine vertebrate species (Foote et al., 2012; Mitchell & Hellberg, 2016) and even identify population-level haplotypes (Baker et al., 2023; Bennington et al., 2024; O'Mahony et al., 2024; Parsons et al., 2025). There is, therefore, enormous potential to apply eDNA metabarcoding for comprehensive biomonitoring of cetaceans; however, quantitative estimates of presence, capture and detection probabilities as a function of sampling and laboratory methods remain largely unexplored. This gap in knowledge has limited the optimization of sampling and processing methods that would enable eDNA to be used as a tool for cetaceans population assessment and management (Suarez-Bregua et al., 2022; Takahashi et al., 2023).

Here, we used data from a highly replicated sampling effort in the CCE to assess the effects of sampling depth on presence, water volume filtered and filtration method on capture, and primer choice on detection of cetaceans in the genus *Delphinus*, of which *D. delphis*

is the only species with two genetically distinct subspecies (*D. d. bairdii* and *D. d. delphis*; long- and short-beaked common dolphins, respectively). We tested two different commonly used mitochondrial primer sets, the MiFish universal primers (Miya et al., 2020), which target fish but also capture cetaceans, and D-loop primers targeting cetacean species (Baker et al., 2018). Using a hierarchical Bayesian occupancy model (Kéry & Schaub, 2012), we show how methodological choices can impact detection frequencies and demonstrate the model's utility in identifying experimental recommendations. While eDNA studies focused on marine biomonitoring are numerous, consistently and accurately capturing rare targets remains a challenge. Our study combines a large, highly replicated set of samples collected and processed using different methods with multi-marker metabarcoding and a flexible Bayesian modelling framework to provide novel guidance on survey design and methodology for eDNA studies of *D. delphis* and other cetaceans. This work demonstrates how existing data can be used to optimize future data collection and has broad implications for improving rare eDNA target detection in the marine environment.

2 | METHODS

2.1 | Sample collection and DNA extraction

Samples used for this project were collected during a California Cooperative Oceanic Fisheries Investigations (CalCOFI) survey in October 2022. While the sampling scheme was designed for a separate study, we leveraged the data to address the questions presented here. Samples were collected at 16 different stations in the southern CCE across varying depths with three different protocols: the NOAA Rockfish Recruitment and Ecosystem Assessment Survey (RREAS), the Global eDNA Marine Collection and Analysis Program (GEMCAP; Dan et al., 2024), and the NOAA-CalCOFI Ocean Genomics Project (NCOG; James et al., 2022) protocols (Figure 1; Table S1). All protocols were used at all stations. The sampling methods targeted similar depths with some variability (e.g. RREAS did not sample below 100 m) and filtered different volumes of water (Figure S1). Sample volume ranged from 1 to 6 L, with GEMCAP averaging approximately 1 L, RREAS approximately 2 L and NCOG approximately 4.5 L per sample (Table S1; Figure S1). All samples were generated with water from Niskin bottles mounted on a CTD rosette filtered through 0.2 µm pore size filters within 1 h of collection. The methods differed in filter material, pump type and DNA extraction method (Figure 2; Table S1). Moreover, the RREAS method included three biological replicates of 2 L each while GEMCAP and NCOG only included a single replicate of 1–6 L (Figure 2). The RREAS and GEMCAP collection protocols are provided in the CalCOFI eDNA Protocols GitHub repository (github.com/calcofi/edna-protocols; DOI: 10.5281/zenodo.14983609). The NCOG method protocol can be found at <https://www.protocols.io/view/noaa-calcofi-ocean-genomics-ncog-sample-collection-eq2lypdorlx9/v1>. At 5 out of 16 stations, a field negative sample was generated following

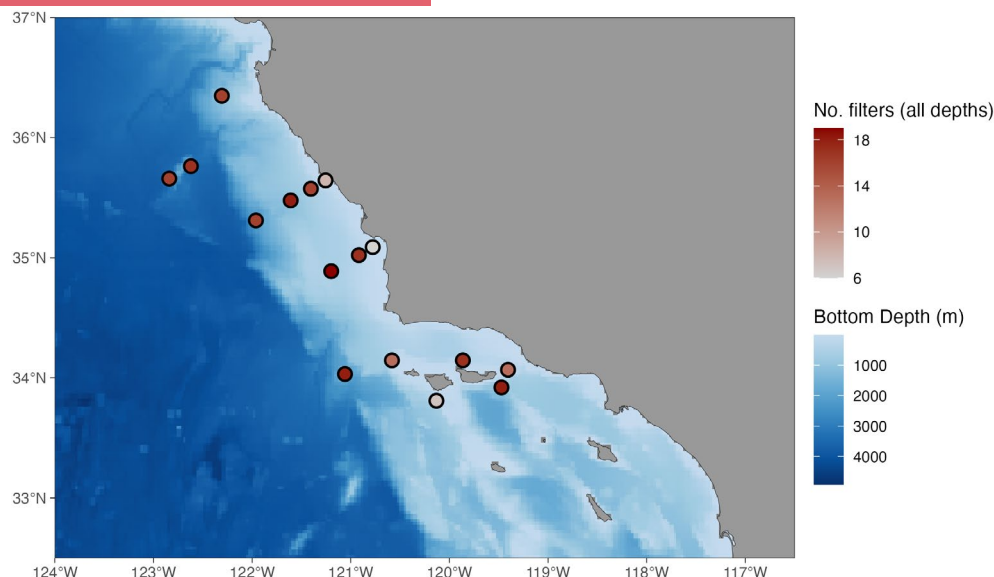


FIGURE 1 Map of sampling sites (stations) and the total number of samples collected at each station at all depth strata. The station bottom depth and number of filters are represented by the associated colour shade.

both RREAS and GEMCAP protocols by filtering 2L (RREAS) or 1L (GEMCAP) milliQ water. All samples were stored at -80°C (GEMCAP and RREAS) or in liquid nitrogen (NCOG) with no preservation buffer until extraction.

All sampling was conducted under the Scientific Collecting Permit ID: S-203450001-21029-003 issued by the California Department of Fish and Wildlife.

DNA was extracted with one of three method-specific protocols: manually using the Omega Bio-Tek Mag-Bind® Blood & Tissue DNA HDQ 96 kit according to the manufacturer's instructions (RREAS); with an automated protocol using the MagMax Microbiome Ultra Nucleic Acid Isolation kit (Applied Biosystems) and a KingFisher Flex (Opentrons Labworks, Inc.) instrument (GEMCAP); and with a mostly automated protocol using a Macherey-Nagel NucleoMag Plant Kit for DNA purification and an Eppendorf EpMotion 5075TMX instrument (NCOG; full protocol: <https://www.protocols.io/view/sterivex-dna-extraction-x54v9m1y4g3e/v2>). Each extraction set included a negative control in which milliQ water was added in place of template DNA. We use the term 'biological replicate' to refer to eDNA samples resulting from individual filters.

2.2 | Amplification and sequencing

eDNA samples were processed through PCR amplification, library preparation and sequencing steps. Amplification was performed using two different primer sets: the universal MiFish primers targeting the 12S rRNA gene (F: GCCGGTAAACTCGTGCCAGC, R: CATAGTGGGGTATCTAATCCAGTTTG; Miya et al. 2015; Sales et al. 2019) and the D-loop region of the mitochondrial control region gene (F: TCACCAAAGCTGRARTTCTA, R: GCGGGTTGCTGGTTTCACG; Dalebout et al. 2004; Baker et al. 2018). MiFish primers are designed to capture 12S rRNA marker gene regions in teleost fish

and elasmobranchs, but also amplify this marker in marine mammals, including cetaceans. Most fish taxa are resolved at the species level with some exceptions (e.g. *Sebastes* spp.; Min et al., 2021), but delphinid marker genes amplified with MiFish primers can often only be resolved to the family or genus level (Valsecchi et al., 2020). In contrast, the D-loop primers used here are specific to cetaceans and can resolve most taxa at the species level (Baker et al., 2006, 2018; Dalebout et al., 2004). Single reactions were run for MiFish amplifications while D-loop reactions were run in triplicate (Figure 1) due to the higher rate of D-loop amplification failure as assessed by gel band visualization (see below). PCR protocols can be found in the CalCOFI eDNA Protocols GitHub repository (<https://github.com/CalCOFI/eDNA-protocols>) under '12S PCR developed for the MURI MMARINeDNA Project' (DOI: 10.5281/zenodo.16729110) and 'MURI D-loop PCR Protocol' (DOI: 10.5281/zenodo.16923327). Negative controls included amplifications of extraction blank samples as well as PCRs with milliQ water added in place of DNA template. Amplification success was checked by running 5 μL of each reaction on a 2.0% agarose gel. D-loop PCRs were considered a success if at least one of the three reactions produced a band. A second round of PCR was performed to append sample-specific indices (with each technical replicate treated as a unique sample) using Illumina Nextera Unique Dual Indices with the following reaction mixture: 12.5 μL Kapa HiFi Hot Start Ready mix, 1.25 μL primer UD, 11.25 μL DNA template + milliQ water (ratio depending on PCR 1 gel band strength; maximum 10 μL template). Amplification was performed using 5 min of initial denaturation, followed by eight cycles of 20s at 98°C , 30s at 56°C and 3 min at 72°C , followed by a final extension of 5 min at 72°C . Amplification success of the second PCR was checked by running 5 μL of each reaction on a 2.0% agarose gel. All samples including negative controls were sequenced even if no gel band was visible; however, band presence and strength were used to normalize reaction volumes added to the pooled library. Each replicate D-loop

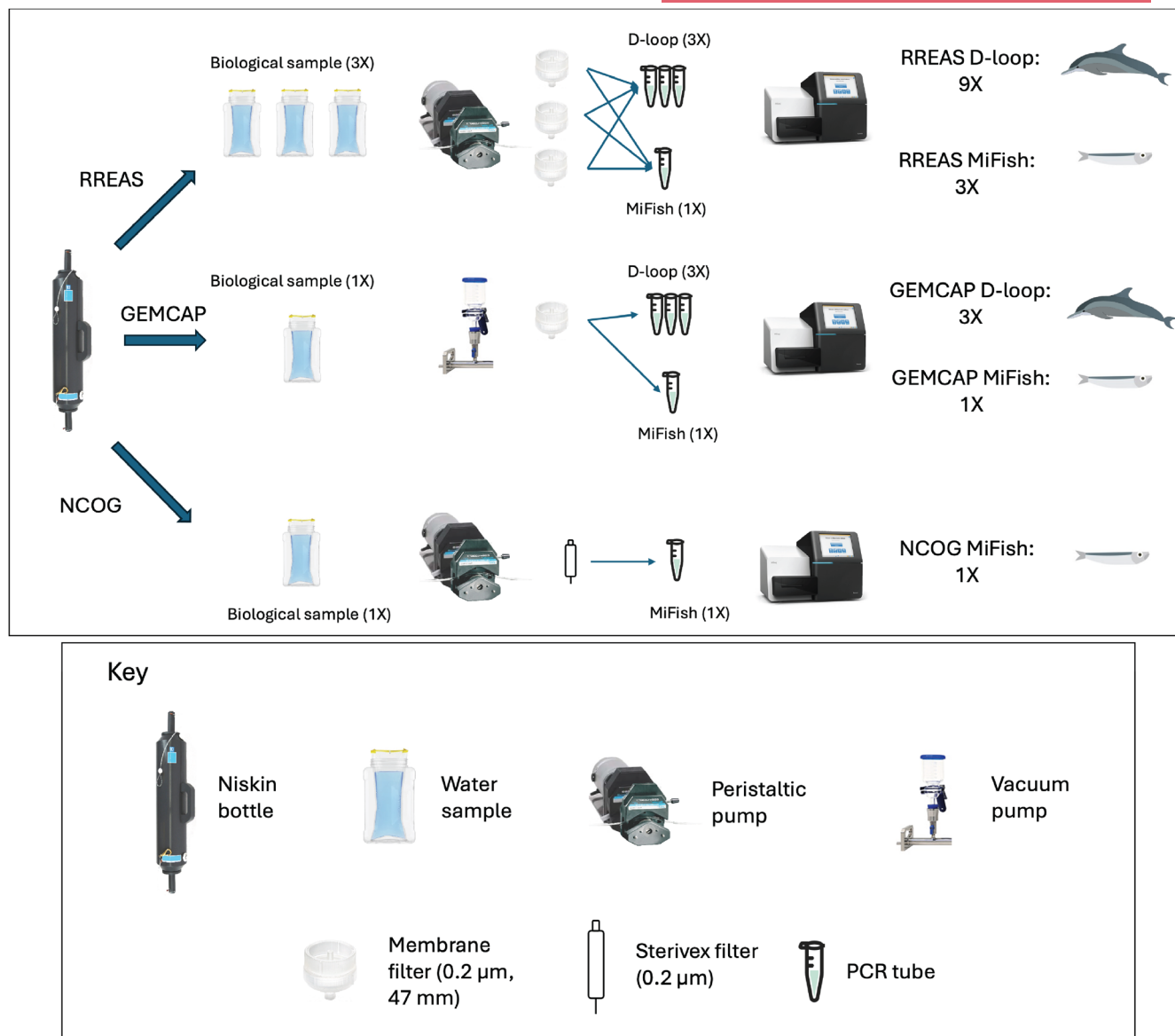


FIGURE 2 Experimental design for each of the three sampling methods: Recruitment and Ecosystem Assessment Survey (RREAS), Global eDNA Marine Collection and Analysis Program (GEMCAP) and NOAA-CalCOFI Ocean Genomics Project (NCOG). Levels of biological and technical replication within each method are denoted by 'X' terminology, for example 3X for three replicates. The Niskin bottle represents one site-depth combination of water, each of which generated multiple filters. Technical replication (PCR) is shown for both primer sets: D-loop (triplicate reactions) and MiFish (single reaction). We use the terms 'biological replicates' to refer to eDNA filters and 'technical replicates' to refer to sequence data samples.

reaction was sequenced independently. Sequencing libraries were pooled and sequenced on an Illumina NovaSeq 6000 instrument at the IGM Genomics Center, University of California, San Diego, La Jolla, CA. We use the term 'technical replicates' to refer to sequence samples resulting from individual PCRs.

2.3 | Bioinformatic processing

Primer sequences were trimmed from the raw sequence data using Cutadapt (Martin 2011), and ASVs were generated with trimmed sequences using DADA2 (Callahan et al. 2016) using the functions *filterAndTrim*, *learnErrors*, *dada*, *mergePairs* and *removeBimeraDenovo*

with default parameters. A length filter was applied to exclude amplicons outside the range of known marker gene lengths (i.e. off-target amplification). For MiFish, the filter excluded amplicons shorter than 163bp and longer than 185bp, and for D-loop the minimum and maximum lengths were 240 and 475bp, respectively.

Sequences were taxonomically annotated with BLAST (Camacho et al., 2009) against the NCBI nt_euk database using the *blastn* task and the following parameters: a minimum percent identity of 97, a word size of 30, minimum *e*-value of $1e^{-30}$, a maximum number of target sequences of 50, a culling limit of three and a taxonomic ID list limiting the search results to vertebrate taxa ('7742.txids' downloaded from NCBI). Raw BLAST output was further processed using a custom script to reconcile conflicting annotations. Briefly,

species were assigned to the highest percent identity with a cut-off of 97%. Taxonomic IDs were linked to the full taxonomy string and run through a Least Common Ancestor analysis with taxonkit (Shen & Ren, 2021) using a modified workflow from Shaffer et al. (2025).

Samples were considered sequencing failures differently depending on the primer set. MiFish primers are much less specific relative to D-loop, which depend on the presence of rare target DNA for detections. Samples were considered sequencing failures and excluded from the model analysis under two circumstances: (a) samples from either primer set with no quality-filtered sequence reads and (b) MiFish samples with no metazoan detections (including fish, i.e. no processed sequence reads) and whose corresponding D-loop samples had no mammal detections. There were no D-loop samples lacking cetacean detections whose corresponding MiFish samples were also empty; thus, most D-loop samples lacking cetacean detections were included in the model.

All bioinformatic scripts are available in the manuscript GitHub repo.

2.4 | Modelling approach

We constructed a hierarchical Bayesian occupancy model to explicitly account for the multiple sources of uncertainty across sampling locations, methods and technical replicates. Our hierarchical model included three levels: occurrence at the depth of each sample, capture in each biological replicate by site, method and volume, conditional on occurrence and detection in each technical replicate by primer, conditional on capture. One of the advantages of this hierarchical modelling structure, where each

biological and technical replicate was its own Bernoulli trial, is that the ability to robustly assess methodological impacts on capture and detection likelihoods was not affected by differences in replication levels among methods. A glossary of model terms is provided in Table 1.

2.5 | Hierarchical model structure

2.5.1 | Presence

Cetaceans are not uniformly distributed in the water column, and therefore, we do not expect their eDNA to be present at all depths. We modelled the presence of cetacean eDNA z at a given sampling location at site s and depth d , $z_{s,d}$, as a Bernoulli random variable with occurrence probability Ψ_d :

$$z_{s,d} \sim \text{Bernoulli}(\Psi_d),$$

where occurrence probability at each depth Ψ_d is a function of sample depth D_d on the logit scale:

$$\text{logit}(\Psi_d) = \beta_0 + \beta_{\text{depth}} \cdot D_d.$$

Here, β_0 is an intercept term and β_{depth} is a covariate for the effect of depth on occurrence probability.

2.5.2 | Capture

Our eDNA monitoring involved 16 fixed sampling sites surveyed at multiple depths using 3 distinct water collection methods, each of

TABLE 1 Glossary of symbols used in hierarchical model.

Symbol	Definition	Type	Prior	Initial value
x_t	Detection in each technical replicate t	Data		
y_b	Capture in each biological replicate b	Variable		x_t
$z_{s,d}$	Occurrence at each site s and depth d	Variable		x_t
Ψ_d	Probability of occurrence at each depth d	Variable		
γ_b	Probability of capture in each biological replicate b	Variable		
D_d	Depth at depth d	Data		
V_b	Volume of biological capture b	Data		
μ	Mean of site-specific capture probability	Parameter	Normal(0, 1.7)	0
σ	Standard deviation of site-specific capture probability	Parameter	Exp(1)	1
β_0	Intercept for occurrence probability	Parameter	Normal(0, 1)	0
β_{depth}	Coefficient for depth	Parameter	Normal(0, 1)	0
β_{volume}	Coefficient for volume	Parameter	Normal(0, 1)	0
δ_{method}	Random effect for method	Parameter	Normal(0, 1.7)	0
Ω_{primer}	Random effect for primer	Parameter	Beta(1, 1)	0.5

Note: The symbol column provides example notation, the definition column explains what each symbol represents in the mathematical model, the type column denotes whether symbols used represent latent model variables, parameters to be estimated or data used to fit the model. For parameters to be estimated by the model, priors and initial values are also provided.

which produced some number of eDNA filters or biological replicates. The presence or absence of cetacean eDNA in each biological replicate y_b was a Bernoulli random variable conditional on the presence of delphinid eDNA at that site and depth $z_{s,d}$ and probability of occurrence in the biological replicate Y_b :

$$y_b \sim \text{Bernoulli}(Y_b \cdot z_{s,d}).$$

We modelled the probability of capture of eDNA in each biological replicate sample Y_b as a function of the site-specific capture probability Φ_s , the volume of sample V_b , and a fixed effect for the filtration method used δ_{method} on the logit scale:

$$\text{logit}(Y_b) = \Phi_s + \beta_{\text{volume}} \cdot V_b + \delta_{\text{method}}.$$

Here, Φ_s was a site-specific random effect with mean μ and standard deviation σ :

$$\Phi_s \sim \text{Normal}(\mu, \sigma)$$

2.5.3 | Detection

Each biological replicate was then subsampled for PCR amplification with two different primer sets. MiFish primers were applied to a single technical replicate and D-loop primers were applied to three technical replicates. We modelled the observed detection of cetacean eDNA in each technical replicate x_t as a Bernoulli random variable with primer-specific detection probability Ω_{primer} conditional on capture in the biological sample y_b :

$$x_t \sim \text{Bernoulli}(\Omega_{\text{primer}} \cdot y_b).$$

The hierarchical nature of the model, conditioning on occupancy at a site and depth, then on capture within a biological replicate, then on detection in each technical replicate, allowed information to be shared across laboratory and sample collection methods, even if a particular method was only applied once per site and depth.

2.6 | Model implementation

We implemented our model in NIMBLE (de Valpine et al., 2017) and ran the model with 3 chains, 200,000 iterations per chain, 10,000 iterations of burn-in and 100-fold thinning. We used uninformative or weakly informative prior distributions for model parameters (Table 1). To assess model convergence, we visually inspected trace plots of model parameters and calculated the potential scale reduction factor (PSRF; Gelman & Rubin, 1992). We considered parameters to have converged when PSRF < 1.1.

2.7 | Data simulation for model validation

Prior to fitting the model to empirical data, we developed a data simulation procedure to validate model performance and verify

parameter identifiability. The simulation code generated synthetic data consistent with our sampling design and hierarchical data-generating process. Specifically, we first specified occurrence probabilities for each unique site-depth combination on the logit scale, incorporating a shared intercept and a fixed effect of depth (centred). For each site-depth combination, we then simulated an occurrence state (occupied or unoccupied). Given occurrence at a site-depth location, we simulated the biological replicate capture process by assigning capture probabilities on the logit scale, incorporating fixed effects of volume filtered (centred) and collection method, as well as a site-specific random effect drawn from a normal distribution. Finally, given capture within a biological replicate, we simulated the detection process at the technical replicate level using primer-specific detection probability parameters. We fit the model to the simulated data and verified parameter recovery by comparing posterior mean estimates and 95% credible intervals to the known true parameter values used in data generation, verifying that the model could reliably recover all key parameters, including presence, capture and detection probability components. The simulation code, along with all data and code to carry out the statistical analyses associated with this research, is available in the provided code and manuscript GitHub repo.

3 | RESULTS

3.1 | Sample collection

Samples were collected from 16 different stations, with a maximum of 19 filters generated from a single station (Figure 1). Stations ranged from very nearshore to 110km offshore. A total of 263 filters were collected, ranging from 800 to 6000mL in water volume and 5 to 515m in depth (Table 2; Figure 3). The maximum number of biological replicates (filters) representing the same site-depth combination was five (three RREAS replicates, one GEMCAP replicate and one NCOG replicate; Table S1). The number of technical replicates (sequence samples) representing the same site-depth combination ranged from 3 to 17, with the maximum number generated from all three sampling methods, both primer sets and associated biological and technical replication (Table S1).

3.2 | Sequence data

A total of 235 filters successfully amplified and sequenced, generating 734 sequence data samples: 570 with the D-loop primers (including technical replicates) and 164 with the MiFish primers. Raw read counts ranged from 546 to 459,406 for MiFish and 1167 to 7,372,033 for D-loop samples. Processed reads ranged from 2 to 194,664 for MiFish samples and 2 to 3,853,568 for D-loop samples. Detection rates of *D. delphis* ssp. from field samples, including all technical replicates, were 9.6% for D-loop and 3% for MiFish primers (Table 1). No D-loop lab negative controls had any cetacean

TABLE 2 Summary of the number of samples collected, sequenced and used in the modelling analysis.

	GEMCAP	RREAS	NCOG	DL:GEM	DL:RREAS	DL total	MFU:GEM	MFU:RREAS	MFU:NCOG	MFU total
Samples collected										
Sample filters	71	123	69							
Sample volume range (mL)	900–2000	800–2000	1800–6000							
Sample volume mean (mL)	1067	1953	4498							
Depth range (m)	5–515	10–100	5–515							
Field blanks	6	5	0							
Samples sequenced										
Biological replicates (filters)				71	120	192	59	117	66	242
Technical replicates (PCRs)				212	358	570	59	117	66	242
Technical replicates failed sequencing				18	0	18	13	50	23	86
% Technical replicates failed				0.08	0	0.03	0.2	0.44	0.35	0.32
Samples modelled										
Biological replicates (filters)				71	120	192	46	67	43	164
Technical replicates (PCRs)				194	358	570	46	67	43	164
Delphinid detection rate				0.06	0.12	0.096	0.02	0.03	0.05	0.03

Abbreviations: DL, D-loop primers; MFU, MiFish Universal primers.

detections. Two D-loop field negative controls had *D. delphis* ssp. detections: one GEMCAP field blank with *D. d. bairdii* (11 reads) and one RREAS field blank with *D. d. delphis* (2 reads). No MiFish field or lab negative controls had any cetacean detections.

A total of 551 technical replicates were amplified and sequenced with the D-loop primers, representing 194 unique eDNA filters, which in turn represented 72 water samples and 113 sample-method combinations. The average percentage of on-target (vertebrate) amplification for the D-loop primer set was 31.2%. Out of these 551 samples, 76 yielded cetacean detections for eight species (Table S4). Two of the species, humpback (*Megaptera novaeangliae*) and sperm whale (*Physeter macrocephalus*), were only found in two samples and one sample, respectively. All other detections were in the family Delphinidae and included short- and long-beaked common dolphin (*D. d. delphis* and *D. d. bairdii*, respectively), bottlenose dolphin (*Tursiops truncatus*) and killer whale (*Orcinus orca*). They also included detections of *D. delphis* that could not be resolved to subspecies.

A total of 156 technical replicates were amplified and sequenced with MiFish primers, representing 68 site-depth combinations and 125 site-depth-sampling method combinations. The average percentage of on-target (vertebrate) amplification for the MiFish primer set was 3.9%. Out of the 156 sequence samples, 5 yielded detections for *D. delphis*. The two subspecies of *D. delphis* present in the CCE could not be resolved using MiFish primers.

The highest number of biological replicates with *D. delphis* ssp. detections from a single site-depth combination was five, from the 5 to 10m depth grouping at one site (Figure S2). The species with the highest number of detections was *D. d. bairdii*, detected in 41 sequence samples with D-loop primers. All other species-level detections occurred in under 10 samples, with *O. orca* and *T. truncatus* only detected in one sample and two samples, respectively.

Given the disproportionate number of *D. delphis* detections in our data set, we chose to limit the modelling input to detections of the genus *Delphinus*, of which *D. delphis* is the only species (*D. d. delphis*, *D. d. bairdii* and *D. delphis* ssp.).

3.3 | Model results

Our model converged, as evidenced by PSRF values <1.1 for all model parameters. Mean and median posterior parameter estimates with 95% credibility intervals (CIs) are given in Table S2.

3.4 | Effect of sampling depth on delphinid eDNA presence

At any given sampling site, the model-predicted median probability of delphinid eDNA presence decreased with water column depth (Figure 4; Figure S2). At 5m, median probability was close to 90%, and at 500m it dropped to approximately 5%.

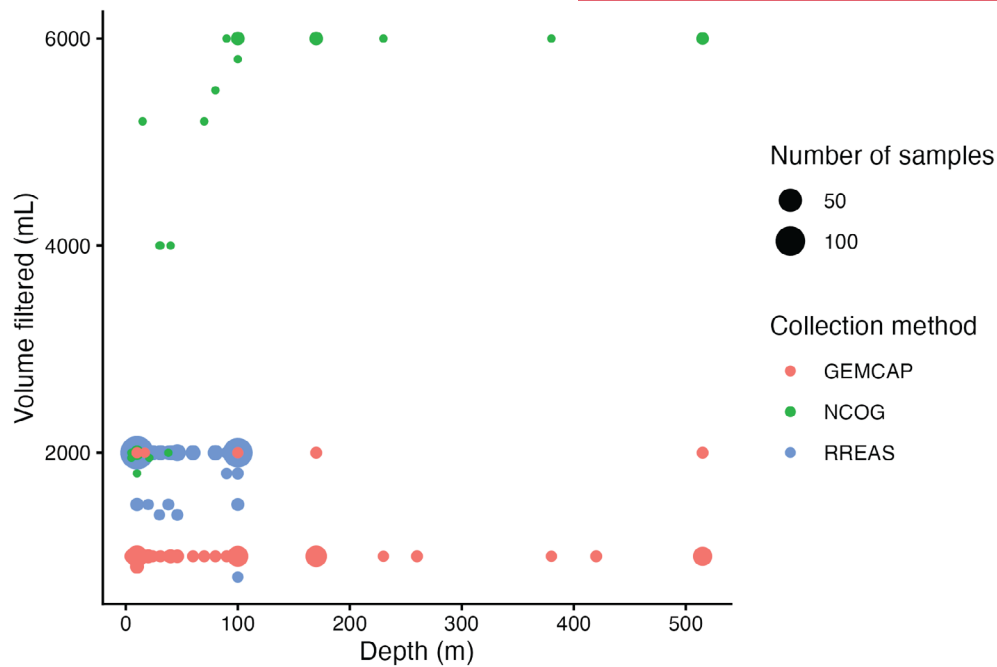


FIGURE 3 Sampling depth and volume filtered for each sample used in this study, coloured by collection method and sized by the number of samples with the specific depth-volume combination.

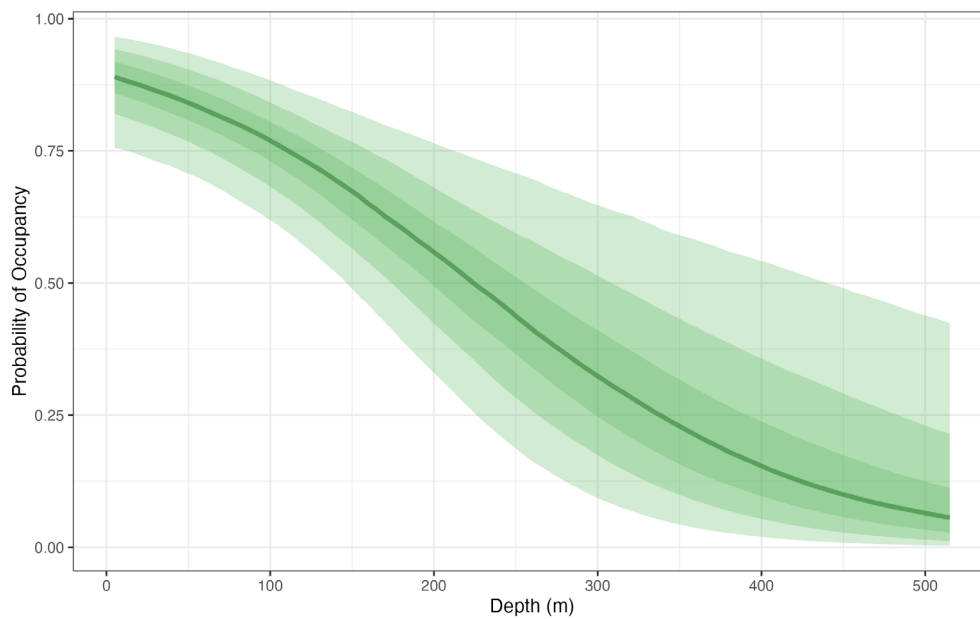


FIGURE 4 Model-predicted relationship between water column depth sampled (horizontal axis) and probability of delphinid eDNA presence (vertical axis). The dark green line indicates the median prediction, with shaded ribbons for the 25%, 50% and 75% posterior credibility intervals. The shallowest 100m featured the highest probability of eDNA presence.

3.5 | Effect of water volume and filtration method on delphinid eDNA capture probability

The volume of water filtered influenced capture probability. Given the presence of delphinid eDNA in a sample, our model predicted that the median posterior probability of capture when filtering 6 L

was close to 100% and when filtering 1 L was approximately 12% (Figure 5; Figure S3). The width of the posterior CIs increased dramatically above 2 L.

Given the presence of delphinid eDNA in a water sample, the estimated probability of capturing it on a filter was similar between the three collection protocols (Figure 6).

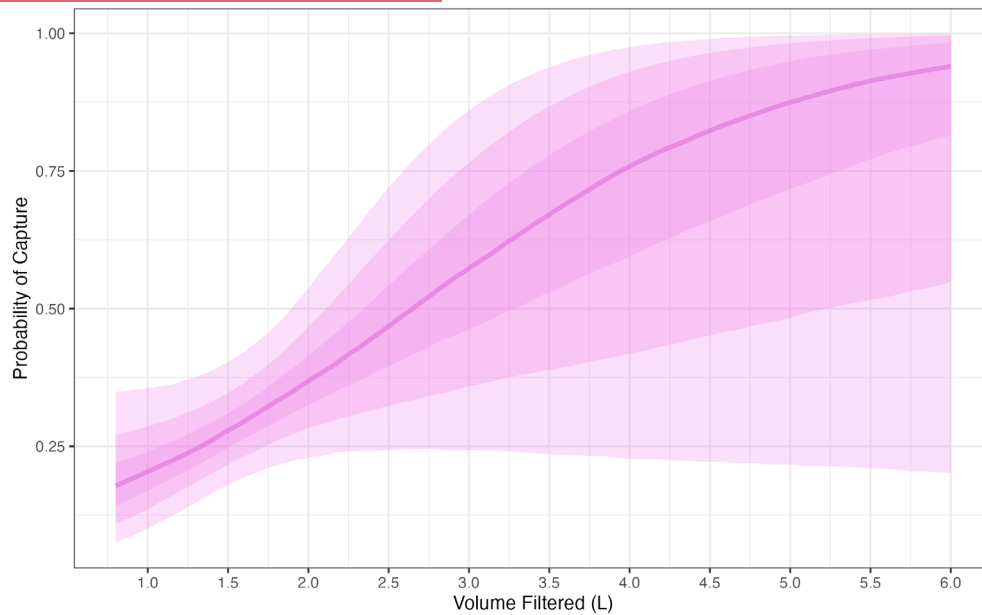


FIGURE 5 Model-predicted relationship between volume of water filtered (horizontal axis) in a sample and the probability of delphinid eDNA capture (vertical axis), given target eDNA presence. The dark pink line indicates the median prediction, with shaded ribbons indicating the 25%, 50% and 75% posterior credibility intervals.

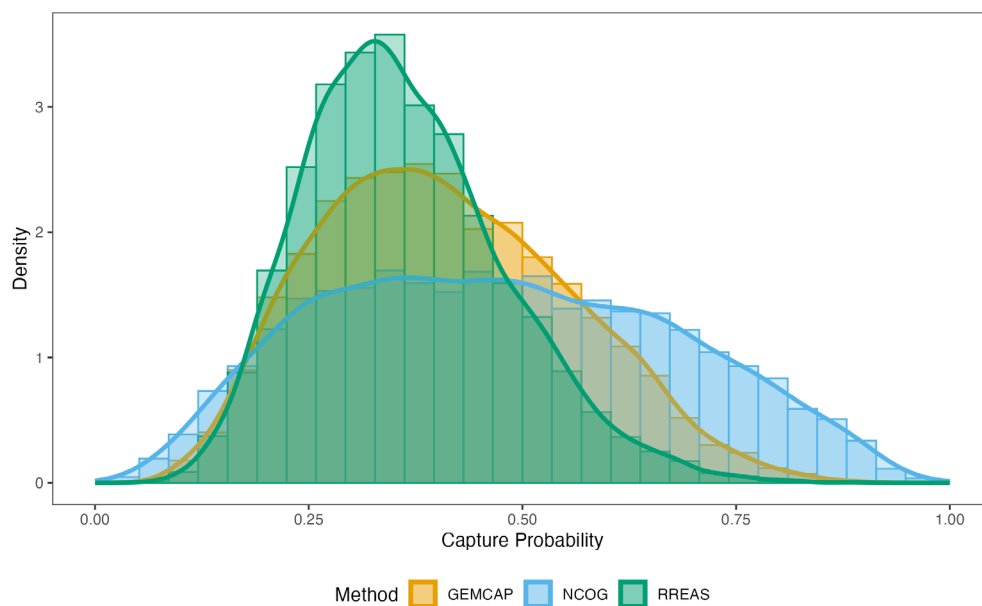


FIGURE 6 Posterior density (vertical axis) of delphinid eDNA capture probability (horizontal axis) for each collection method (colours) shows RREAS had the best constrained posterior density, suggesting capture rate was less variable than the other two methods.

3.6 | Effect of primer set on detection probability

Assuming the target molecules were captured on a given filter, primer choice strongly affected detection probability in the sequencing data. The D-loop primer set was much more likely to generate delphinid detections relative to the MiFish primer set (Figure 7).

4 | DISCUSSION

Capturing rare targets in marine eDNA is a formidable challenge due to limits of molecular biology and poorly constrained rates of shedding, generation, dispersal and decay, genetic variability, and other factors yet to be delineated (Ficetola et al., 2015; Harrison et al.

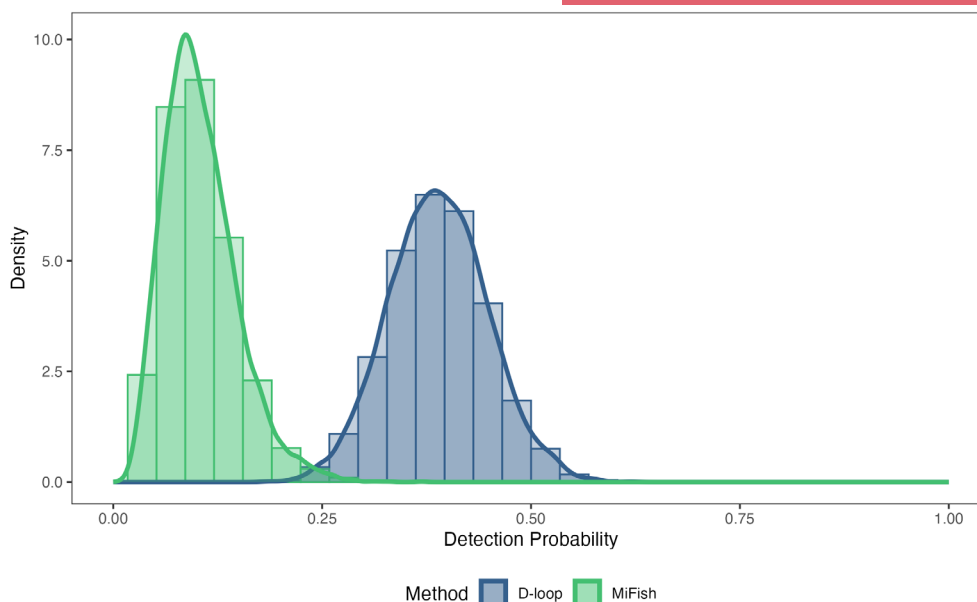


FIGURE 7 Posterior density (vertical axis) of delphinid eDNA detection probability (horizontal axis) for each primer set shows the median posterior density for D-loop was 36.4% compared with 8.6% for the MiFish primers.

2019; Gold et al., 2023). Our study presents a systematic approach to improving the likelihood of rare eDNA target capture and detection by employing occupancy modelling techniques (MacKenzie et al., 2002; Schmidt et al., 2013). We used a hierarchical Bayesian occupancy model (Kéry & Schaub, 2012) to separate the effects of target molecule presence, capture and detection on eDNA observations associated with an intensive, ship-based eDNA sampling survey in the CCE. Similar models have previously been used in eDNA studies, particularly in the freshwater and terrestrial fields (e.g. Schmidt et al., 2013; Dorazio & Erickson, 2018; McColl-Gausden et al., 2024; Tomke & Price, 2025), as well as in marine systems (e.g. McClenaghan et al., 2020; Valdivia-Carrillo et al., 2021).

Cetacean eDNA is an extremely rare molecular target but also offers one of the most potentially impactful uses of eDNA due to the combination of high interest for conservation and regulation and the limited understanding of cetacean distribution and abundance, particularly below the ocean surface and in offshore habitats. Because *D. delphis* ssp. are among the most common cetaceans in the Southern CCE, they provide an ideal case study species to use when testing the effect of sample collection and processing parameters on the probability of detecting rare species. By limiting our modelling analysis to *D. delphis* ssp., which have highly similar life strategies and behaviour, we also avoided conflating data from species that may have different depth distributions. Our results indicate that the likelihood of detecting rare eDNA targets can be greatly improved by (a) sampling larger water volumes, (b) sampling from the depth stratum most appropriate for the target organism and (c) using taxon-specific primers.

The results of our analysis revealed a notable finding: at any given site in our survey, the model-estimated presence of delphinid eDNA at the surface is above 90% (Figure 4), suggesting that delphinid eDNA molecules are widespread within our study area. This model outcome is reflected in our empirical delphinid detection rates,

which were 9.6% for D-loop and 3% for MiFish primers (Table 2). However, D-loop samples from the shallowest part of the water column (5–10m) had a detection rate of 13.5%, while the rate for D-loop samples taken from below 100m was only 0.35%. The discrepancy between predicted occupancy and detection rates reflects the importance of sampling and processing methodology. At any given location, delphinid molecules are likely to compose a small fraction of the total eDNA pool, and thus, they will not consistently be detected by standard sampling and processing methods even when present. Samples amplified with the MiFish primer set only had a 4.3% detection rate even from samples taken at or above 10m, and yielded no detections below 100m. The difference between primer sets was also notable in the amount of off-target amplification, which was 31.2% for D-loop but 96.1% for MiFish primers, reflecting the challenge of detecting rare molecules even with highly targeted assays. While D-loop primers improved detection probabilities over MiFish primers (Figure 7), further refinement and optimization of molecular methods, including in primer choice, may continue to improve detection rates and provide more informative study outcomes.

The high estimated probability of delphinid eDNA occurrence is reasonable in a molecular context. Common dolphins (*D. delphis*) are the most frequently sighted cetaceans on CalCOFI surveys (Campbell et al., 2015). They have a patchy distribution, with individuals travelling in groups ranging in size from hundreds to thousands. These groups likely generate large depositions of eDNA; additionally, dolphins travelling in these large groups may shed more eDNA due to frequent contact and rubbing between individuals. eDNA can remain at the surface for up to 48h (Brandão-Dias et al., 2025; Collins et al., 2018), although decay rates likely follow a biphasic model (Andruszkiewicz Allan et al., 2021) with the majority of the material degrading within the first 6–24h. eDNA can also be transported short distances with ocean currents (Allan et al., 2021; Ellis

et al., 2022; Jo et al., 2025; Shea et al., 2022); while limited, these processes may nevertheless contribute to widespread molecule distribution, particularly in a region where the source organism has a relatively high density. Our findings further reflect those of Nikolic et al. (2025), who described their surface ocean detections of delphinids in a marine mammal metabarcoding survey as 'almost ubiquitous'. It is, therefore, reasonable to conclude that delphinid eDNA will be widespread and can be consistently captured and used to assess species- or subspecies-level distributions.

The apparent ubiquity of delphinid eDNA in the surface ocean is both a challenge and an opportunity for applying eDNA data to delphinid management and conservation. Species distribution models (SDMs), which form the basis for most marine mammal management policy, rely on differential detection rates to encode spatial variability in species presence; without differential detection rates across space, only uniform distributions can be inferred. Our estimated likelihood of presence applies to detections aggregated for both detected *D. delphis* subspecies; however, subspecies-specific detection rates were more nuanced, and distributions of individual subspecies were not uniform across our sampling grid. Thus, for species or subspecies with clear genetic delineations, our sampling and analysis framework would provide an informative data stream for SDMs. Moreover, due to unknown factors like rates of amplification bias and the relationship between template DNA concentration and animal abundance, metabarcoding data are currently limited to presence/absence; however, more quantitative approaches, such as species-specific qPCR assays (Lacoursière-Roussel, Côté, et al., 2016; Salter et al., 2019) or metabarcoding in concert with Universal Molecular Identifiers (Fields et al., 2021; Lin et al., 2024) can provide relative abundance information, which would better inform distribution models. These species-specific assays may also be more appropriate for extremely rare organisms, whose signal can be lost in a multi-species assay like metabarcoding.

Importantly, our results provide guidance for optimizing sampling strategies for cetaceans more broadly as well as for other rare marine species. We not only provide direct method comparisons showing the differences in detection variability, but also identify the key methodological elements likely needed to maximize detections. For example, increasing sample volume improves rare target detection, with detection probabilities trending upward with higher volumes (Figure 5). Only 4.1% of our samples were generated with >2 L of water; most of those samples originated from water depths greater than 50 m, where occurrence probabilities were low, and none of them were amplified with the D-loop primers, which greatly affected detection probability. The CIs of our model were therefore quite wide beyond 2–3 L (Figure 5); we hypothesize that if 5 L or more were consistently filtered, our detection rates of delphinids and other cetaceans would greatly increase. These findings may be related to the patchiness of rare target molecules, which is reflected in other studies showing high levels of false negative cetacean eDNA detections even when sampling in close proximity to animals (Pinfield et al., 2019; Robinson et al., 2023; Székely et al., 2021) and the benefits of large-volume sampling for metazoan eDNA detection

more generally (Govindarajan et al., 2022). If spatial patchiness is indeed a strong contributing factor, then increasing the spatial scale or resolution of the sampling approach would further improve the likelihood of detection. This could be done either by aggregating multiple small-volume bottle samples taken along a transect, or by continuous surface sampling using the underway water system present on most oceangoing vessels. The latter approach has the added advantage of being relatively accessible relative to Niskin bottle sampling, which requires specialized equipment (Boyse et al., 2023; Valsecchi et al., 2021). However, we note that increasing detection probability by aggregating spatially will decrease spatial resolution, which is one of the potential strengths of eDNA sampling methods. Moreover, the risk of cross-sample contamination from underway systems may be elevated relative to discrete bottle-based sampling. While additional work is needed to identify the effect of patchiness on detection likelihood, our study shows that large-volume samples provide the highest rates of detection per sample and is thus the best allocation of sampling resources for cetacean biomonitoring.

Evaluation of primer choice in delphinid detection frequency and specificity provided valuable information on the most appropriate assay for our study. In particular, the use of the D-loop primer set offered a clear advantage for delphinid detection compared with the MiFish primers. This result is likely due to the greater specificity of D-loop primers, whose off-target amplification rates were much lower relative to the MiFish primers (31.2% and 96.1%, respectively). Thus, the signal of delphinid eDNA is less likely to be lost in reads resulting from off-target amplification and the competition effects inherent to the compositional nature of metabarcoding (Gloor et al., 2017; Shaffer et al., 2026). These results reflect the findings by both Deng et al. (2025) and Nikolic et al. (2025) that general vertebrate primers were unsuccessful at capturing cetaceans and that bespoke, taxon-specific primer design was needed for survey success. Moreover, the D-loop target region used in this study can resolve cetaceans to the species and often subspecies level, a key advantage in the potential application of eDNA results to SDMs. However, neither primer set provided consistent detections of baleen whales, which should be captured by both primer sets. We do not yet know whether our eDNA detections of baleen whales accurately reflect their density in the region, or whether they reflect a failure of eDNA capture or detection. This uncertainty highlights an important methodological gap that needs to be overcome for comprehensive eDNA-based cetacean biomonitoring.

The results from our cetacean-specific (D-loop) primers provided an opportunity to explore knowledge gaps in delphinid genetics and consider new opportunities for eDNA-based ecological studies. In the Southern CCE (23°N–34.5°N and from the coast out to 132.5°W), short-beaked common dolphins (*D. d. delphis*) have an estimated density of 0.412 per km² (Becker et al., 2022), while long-beaked common dolphins (*D. d. bairdii*) are less abundant with an estimated density of 0.132 per km² (Becker et al., 2022). Our study area overlapped with the northern part of this range, where short-beaked common dolphins are expected to be even more abundant relative to long-beaked common dolphins (Becker et al., 2022; Campbell

et al., 2015; Douglas et al., 2014). Yet, *D. d. bairdii* was the most frequently detected cetacean subspecies in our eDNA data set (37 out of 551 D-loop samples, compared with 7 samples with *D. d. delphis* detections). While primer bias can certainly affect detection rates of different species, in this case the D-loop primers match to both *D. delphis* subspecies gene sequences 100% and thus should not have been a factor in biased observation. One possible explanation for this discrepancy is that many of the unknown sequences currently assigned to *D. delphis* ssp. (21 D-loop samples) belong to *D. d. delphis*. The D-loop target gene region has sufficient variability to distinguish the two subspecies; however, the full range of genetic diversity may not be represented in the reference database (NCBI nt_euk as of January 2025), thus preventing confident assignment of novel sequences to the subspecies level. The challenge of incomplete reference databases applies to the entire eDNA field, with many species only represented by one or a few reference sequences from voucher specimens (Ardura, 2019; Hestetun et al., 2020; Keck et al., 2023; Zhou et al., 2025) and potential population-level diversity yet to be uncovered. Alternatively, oceanographic conditions related to a changing climate may be leading to range shifts for this species, as an increased occurrence of *D. d. bairdii* along the California coast has been observed during warm (El Niño) water conditions (Carretta et al., 2011). Visual survey data are often pooled across multiple years to estimate density and abundance; a potential advantage of eDNA sampling is the ability to link occurrence to oceanographic phenomena at finer temporal and spatial scales. The differences in detection frequencies for these subspecies between survey methods suggest that eDNA could provide an additional, complementary data stream to visual surveys, helping to better inform estimated distributions of more cryptic cetaceans.

Given the results of our study, we recommend the following work flow for anyone seeking to perform eDNA based surveys of rare molecular targets: first, it is critical to define the research goals and, in particular, assess the need for quantitative versus presence/absence data. If target organism abundance is necessary for downstream decision making, then semi-quantitative approaches like qPCR should be prioritized over metabarcoding. Second, in either case, an appropriate marker gene should be selected based on the primer selectivity (for species-specific assays) or taxonomic resolution (for metabarcoding), as well as its prevalence in public sequence databases. Ideally, multiple well-curated reference sequences for each target species will ensure unambiguous taxonomic assignment. Third, a pilot sampling experiment coupled with an occupancy model should be performed to establish best practices for consistent and reliable detections of rare marine taxa.

This last component of our recommendations is critical, as we note that our results are ecosystem-specific and depend on the biology and ecology of delphinid populations in the given study area. Performing a similar assessment of depth, replication and assay choice on target species detection probabilities before designing any biomonitoring programme, particularly for rare targets, would provide tailored guidance to improve survey design in different regions. As mentioned above, such a study could be expanded to

include molecular assay variables, whose design can play an important role in detection frequencies (e.g. Székely et al., 2021; Alter et al., 2022; Nikolic et al., 2025; Deng et al. 2025). We also assumed homogeneity of detection and capture probabilities conditional on the variables in our model. This may be a simplification; other factors could impact capture and detection probabilities, including environmental variables like water temperature or patchy distribution of target molecules in the water column. The random effect for site may help account for some of these sources of variability in capture, since we assume that most environmental covariates would apply at the site level. A more complex model could include oceanographic variables directly (e.g. Becker et al., 2022), or use geographic proximity to infer that some sites should be more similar than others, for example via a smooth on latitude and longitude or water column depth. Additionally, we did not explicitly test many filtration or laboratory practices that can affect detection rates, such as filter pore size or PCR protocols, which are addressed in other studies (e.g. Lacoursière-Roussel, Rosabal, & Bernatchez, 2016; Deiner et al., 2018; Van Der Loos & Nijland, 2021; Shaffer et al., 2025). We recommend applying a model similar to ours to identify the key variables affecting sampling success and tailoring them to the study design and available resources specific to each programme.

eDNA monitoring represents a transformative tool in marine conservation, capable of providing detailed insight into the distribution and abundance of rare organisms like marine mammals (Taberlet et al., 2018). The development of more accurate SDMs would benefit from the scale and specificity provided by eDNA detections, particularly for species that are difficult to differentiate visually. Our research demonstrates the power of integrating eDNA analysis with sophisticated statistical modelling, paving the way for more comprehensive and accurate assessments of marine ecosystems (Lacoursière-Roussel, Côté, et al., 2016; Rodríguez-Ezpeleta et al., 2021). By providing quantitative estimates of occurrence and detection probabilities, our study not only enhances the reliability of eDNA-based delphinid monitoring but also establishes a strong case for its widespread adoption in marine conservation efforts (Bohmann et al., 2014). The near-ubiquitous presence of detectable eDNA signals across sampling sites indicates that this method could both simplify and improve our ability to monitor and protect marine mammal populations, including those that are traditionally difficult to observe. However, improvements in DNA reference databases remain a limiting factor for consistent species- and subspecies-level differentiation, particularly for certain commonly used marker genes. Moreover, detections of the rarest cetaceans would benefit from rigorous molecular assay optimization to generate the data quality necessary for robust SDMs. In particular, improving the quantitative interpretations from marine mammal eDNA data (e.g. estimated group sizes or biomass) would integrate well into existing SDM frameworks. The modelling framework presented here offers a robust scaffolding for continued development of innovative sampling and processing techniques, potentially reshaping our approach to marine biodiversity assessment and conservation planning.

AUTHOR CONTRIBUTIONS

Nastassia V. Patin: Conceptualization; data curation; formal analysis; investigation; methodology; visualization; writing—original draft. **Eiren K. Jacobson:** Methodology; validation; visualization; writing—original draft; writing—review and editing. **Zachary Gold:** Conceptualization; investigation; methodology; writing—review and editing. **Julie Dinasquet** and **Robert H. Lampe:** Methodology; investigation; writing—review and editing. **Amy M. Van Cise:** Writing—review and editing. **Bryce Ellman:** Methodology; writing—review and editing. **Anne Schulberg:** Methodology. **Noelle Bowlin** and **Ryan Freedman:** Funding acquisition; investigation. **Erin Satterthwaite:** Writing—review and editing. **Andrew E. Allen:** Methodology; funding acquisition. **Brice X. Semmens:** Conceptualization; formal analysis; funding acquisition; writing—original draft.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest to declare.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210x.70370>.

DATA AVAILABILITY STATEMENT

All raw sequence data are available on the NCBI Sequence Read Archive under BioProject ID PRJNA1322596. All code used for data analysis is available on the manuscript GitHub repository: http://github.com/CalCOFI/eDNA_patch and is archived on Zenodo (<https://doi.org/10.5281/zenodo.20834452>; Semmens et al., 2026).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Location and number of delphinid detections, presented by sample depth grouping.

Figure S2. Model-predicted relationship between water column depth sampled (horizontal axis) and probability of delphinid eDNA presence (vertical axis).

Figure S3. Model-predicted relationship between volume of water filtered (horizontal axis) in a sample and the probability of delphinid eDNA capture (vertical axis), assuming eDNA presence.

Table S1. Table of samples used in this study with associated collection data and marine mammal detections.

Table S2. Mean and median posterior parameter estimates with 95% credibility intervals (CIs) for the hierarchical model run with study data.

Table S3. Glossary of symbols used in hierarchical model.

Table S4. All cetacean detections in all sequenced samples, including number of reads matching each annotation.

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